WORKSHEET – HYPOTHESIS TESTING

1. It is claimed that the average life of a particular brand of tire is at least 30,000 km. A travel company suspects this claim. For this reason, they test the average life of 45 tires and determine that their mean is 27400 km, and their standard deviation is 1251 km. According to this result, will they be right in their suspect? (α=0.01)

(Ans: $z_0 = -13.94$, they are right in their suspect)

2. According to the population census made five years ago, 41% of the families living below the poverty line were determined. To investigate whether this proportion has changed, 181 of 400 randomly selected families were found to live below the poverty line. Can this be significant statistical proof that there is an increase in the proportion of families living below the poverty line from the last population census to the present? (α=0.05)

(Ans: $z_0 = 1.73$, yes, it can be significant statistical proof)

3. It is observed that the weight of loaves of bread produced by a bakery is very varied. The owner of the bakery claims that the standard deviation of bread weights is 20 gr. In order to investigate this claim, the weights of 12 loaves of bread are measured and the standard deviation is calculated as 33. Does this result refute the claim of the baker for α=0.01?

(Ans: $\chi_0^2 = 29.95$, yes, it refutes the claim of the baker)

4. To compare the average height of one-year-old male babies in two countries, a sample is randomly selected from each country and the following results are obtained:

$n_1 = 130 \qquad \overline{X_1} = 73.6 \qquad s_1 = 2.46$

$n_2 = 140 \qquad \overline{X_2} = 72.8 \qquad s_2 = 2.34$

Test the hypothesis “there is no significant difference between population means” with the opposite hypothesis for α=0.01

(Ans: $z_0 = 2.73$, there is significant difference)

5. To measure the effectiveness of a painkiller, 200 volunteers are tested. 120 of them are given the capsules containing the actual drug and 80 of them are given the capsules containing the powdered sugar. If 62 people from the real medicinal group and 34 people from the powdered sugar group say they see the benefit of the medicine, what can we conclude about the effectiveness of the painkiller? (α=0.05)

(Ans: $z_0 = 1.25$, the painkiller is not effective)

6. It is claimed that the variance related to the height of the women living in Sanliurfa is larger than that of women living in İzmit. A sample of size 25 has been selected from Sanliurfa and another sample of size 31 has been selected from İzmit. If their variation is respectively 16 and 8, is there significant proof for the claim? (α=0.01)

(Ans: $F_0 = 2$, there is no a significant proof for the claim)

Solve the same problem for the claim “there is a significant difference between population variances” for the problem.

(Ans: $F_0 = 2$, there is no a significant proof for the claim)

7. It is claimed that the average monthly cultural expenditure of university graduates working in the private sector is more than 60 TL. The following data were computed for a randomly selected sample of 9 units: {90, 50, 40, 120, 115, 80, 65, 70, 80} If the population has a normal distribution and $\sigma^2 = 225$, what would your decision be at significance level α=0.05? Solve the problem by computing i) the test statistic, ii) critical value for the sample mean and iii) p value. Also, compute the power of test if the real mean μ is 75.

(Ans: i) $z_0 = 3.78$, there is significant proof to accept the claim, ii) critical value for the sample mean is 68.23. Since $\bar{x} = 78.88 > 68.23$, there is significant proof to accept the claim, $p \cong 0.00$, there is significant proof to accept the claim, $1 - \beta = 0.91$)